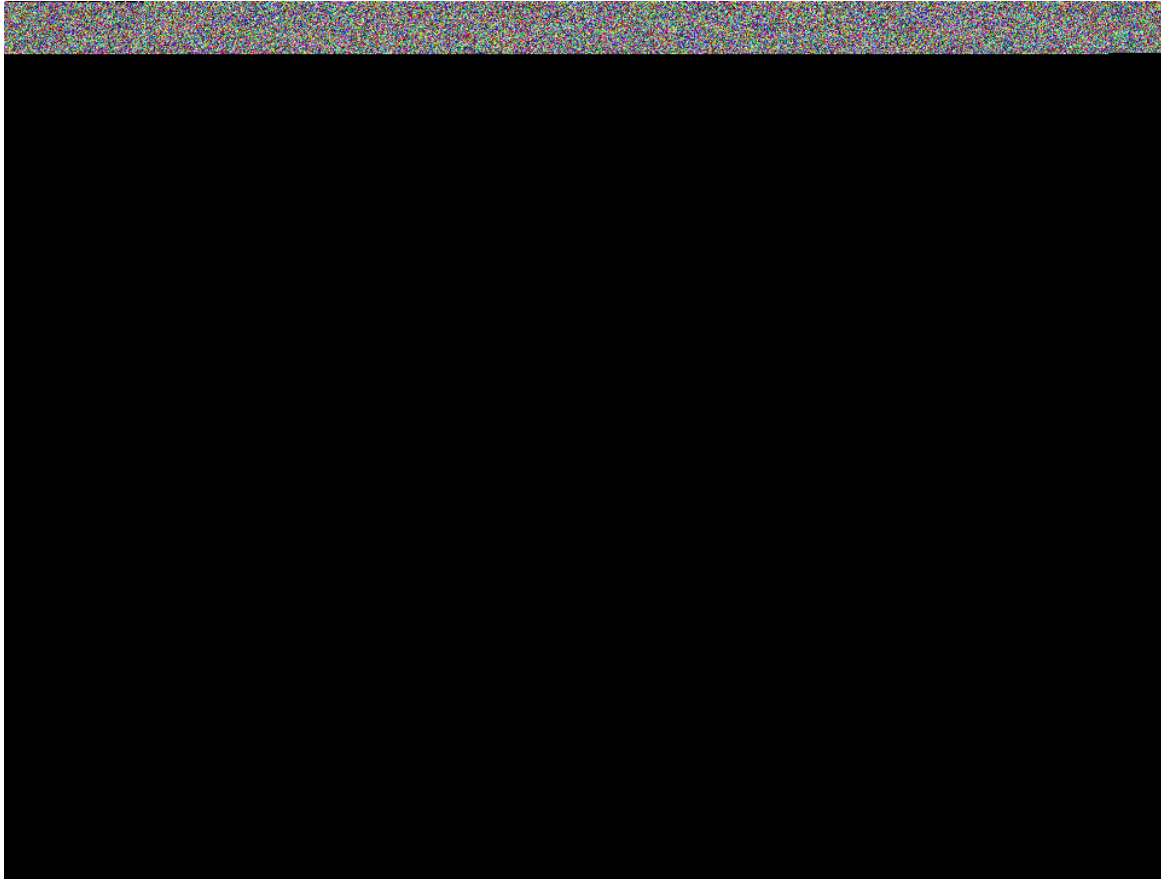


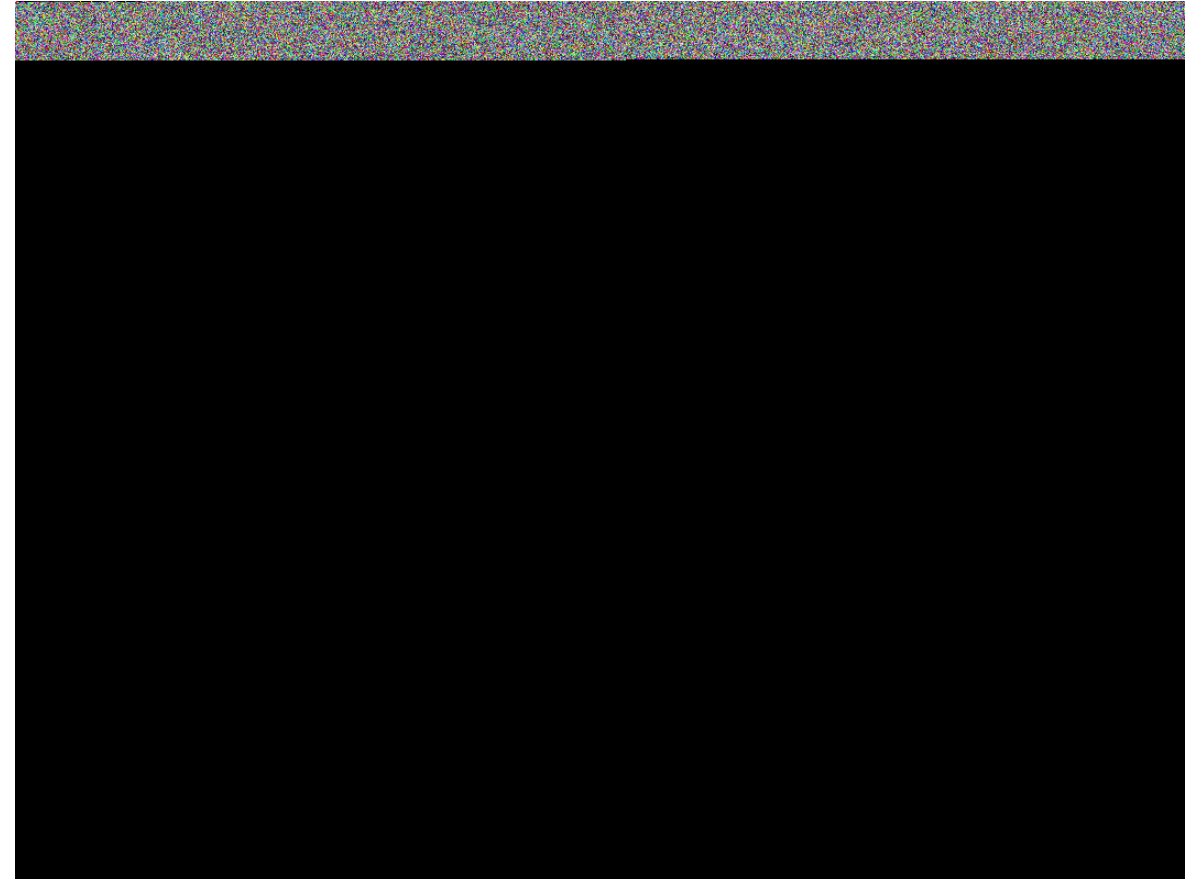
Mechanisms of Toughening in TRC materials

Barzin Mobasher, Arizona State University

Design at the microstructure level will have a profound impact at the macro-structural level strength and durability



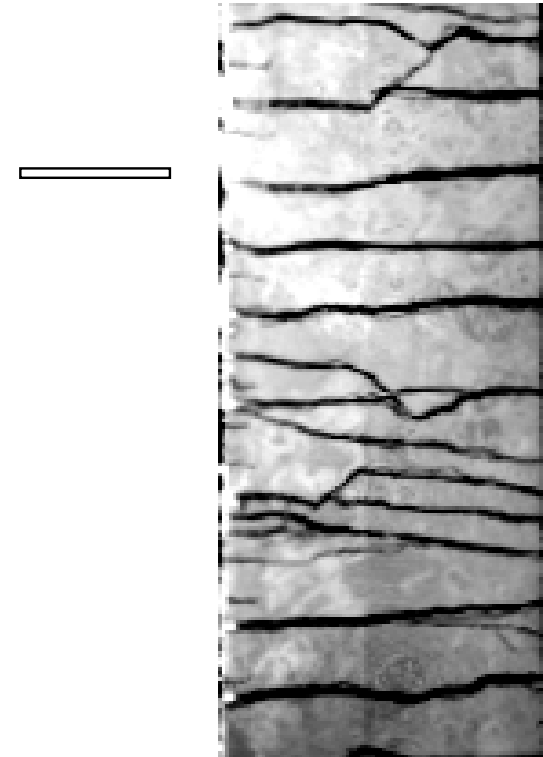
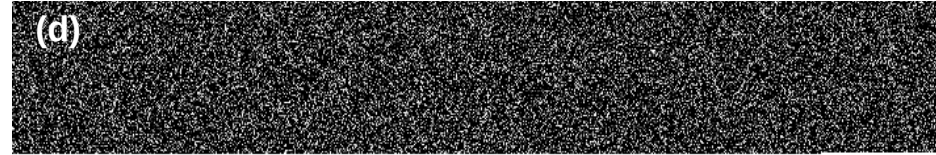
Ordinary reinforced concrete slab, 10" thick,
standard rebars, 18" apart



Fiber reinforced Concrete, reinforced at the microstructure
with $\frac{1}{2}$ " long fibers that are millimeters apart

Characteristics of Toughening at a microstructural level

- Improved Ductility, Toughening, tensile strength
- Increase level of energy absorption
- Fatigue life, impact/explosive loading
- Seismic resistance
- Reduced labor, construction time, scheduling
- Corrosion damage mitigation
- Long-term repair and maintenance.

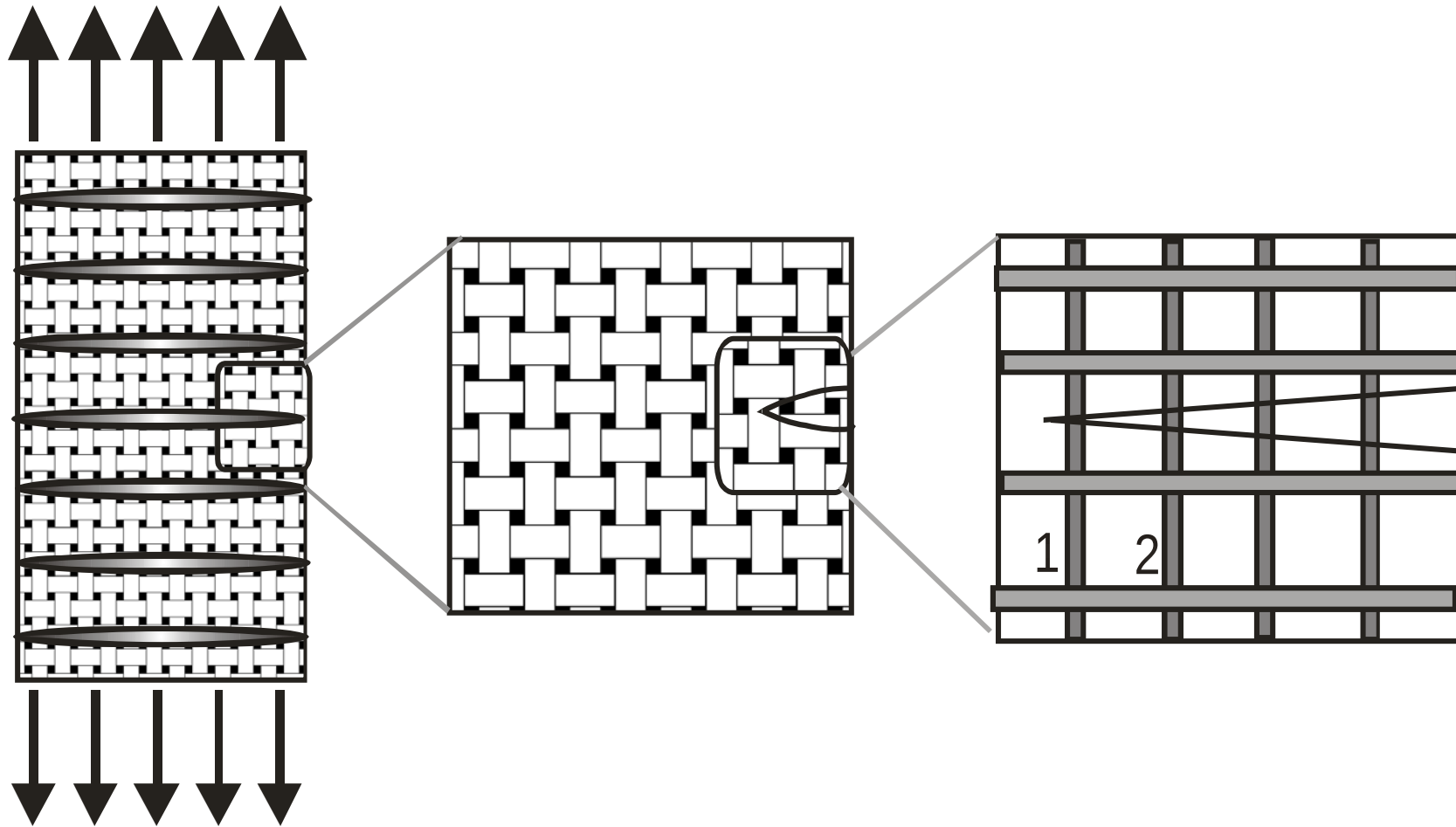


Applications for UHPC Bridge Applications

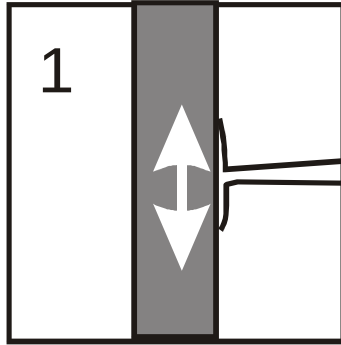


ADOT SPR 745 Project

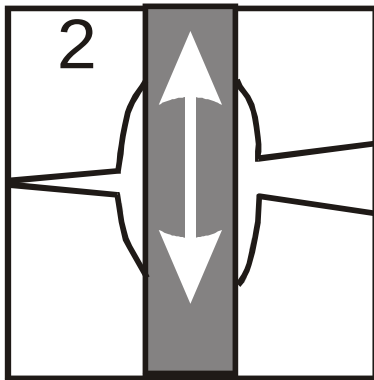
Suppression of Crack growth by means of Fibers



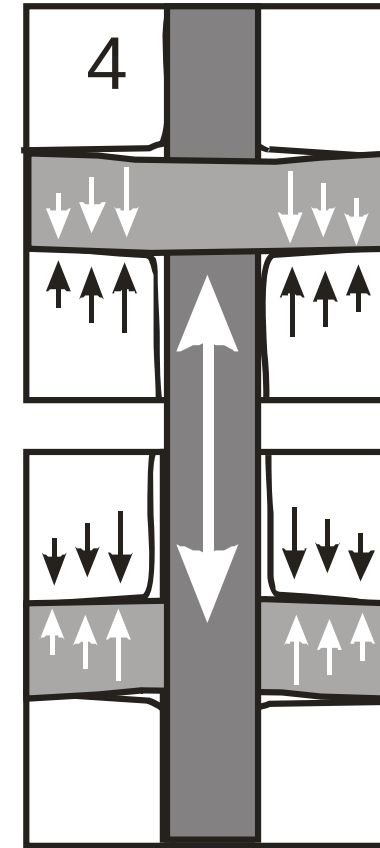
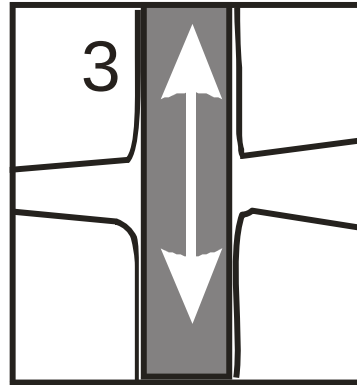
4 Stages of Fiber Bridging Mechanisms



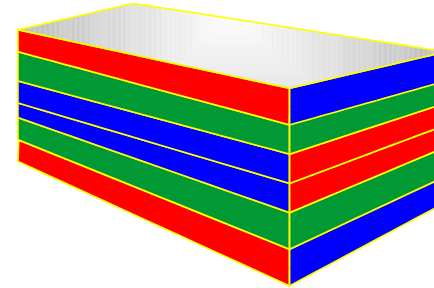
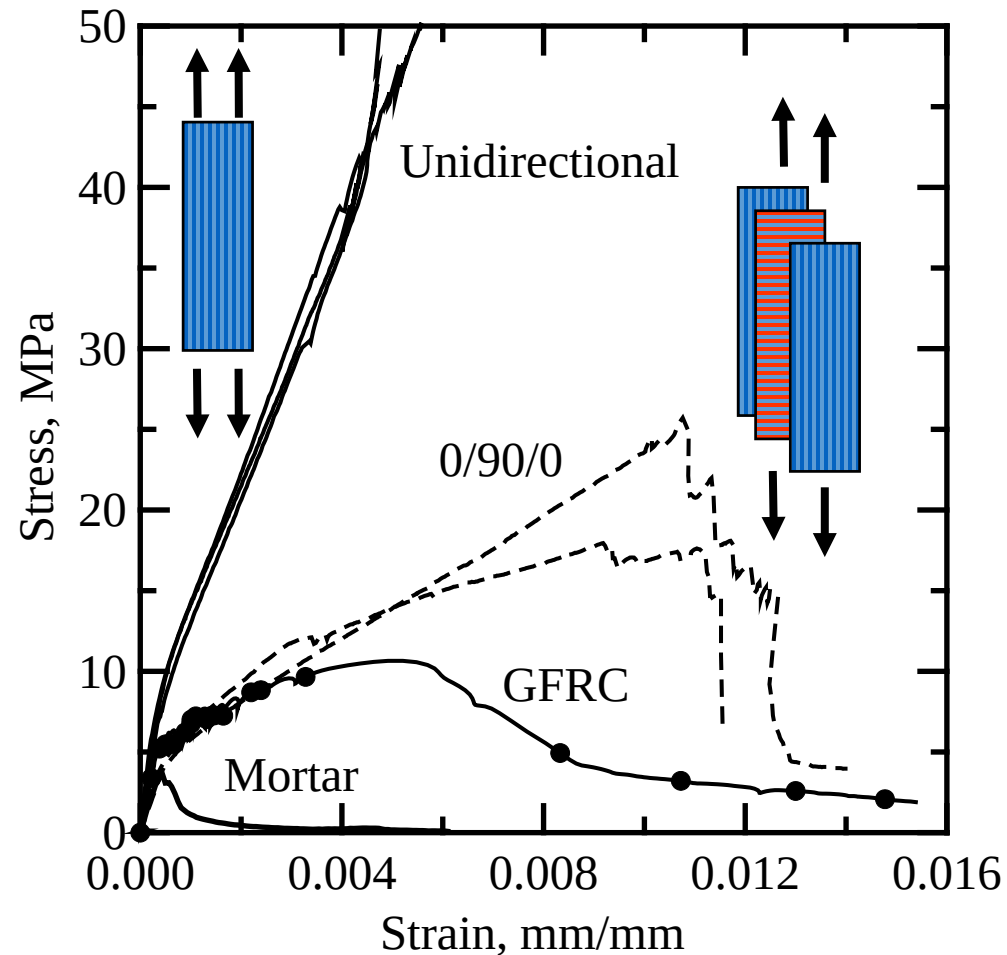
Crack Deflection



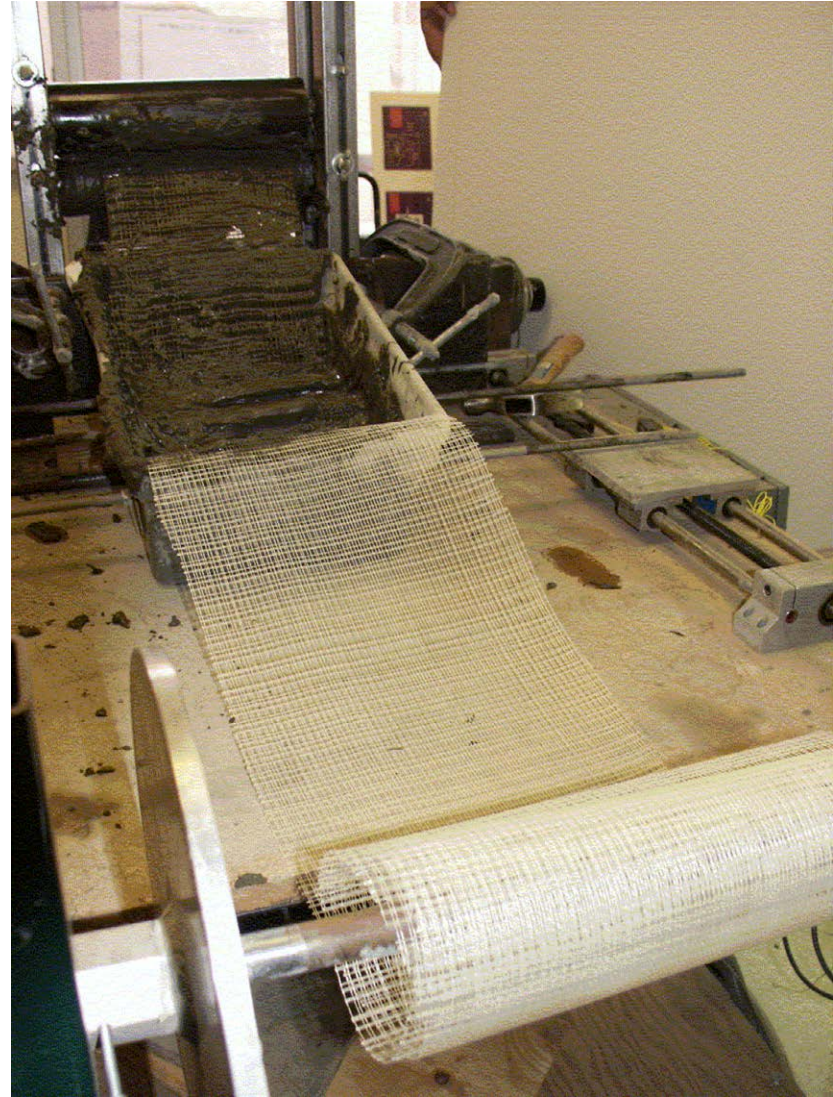
Yam Debonding



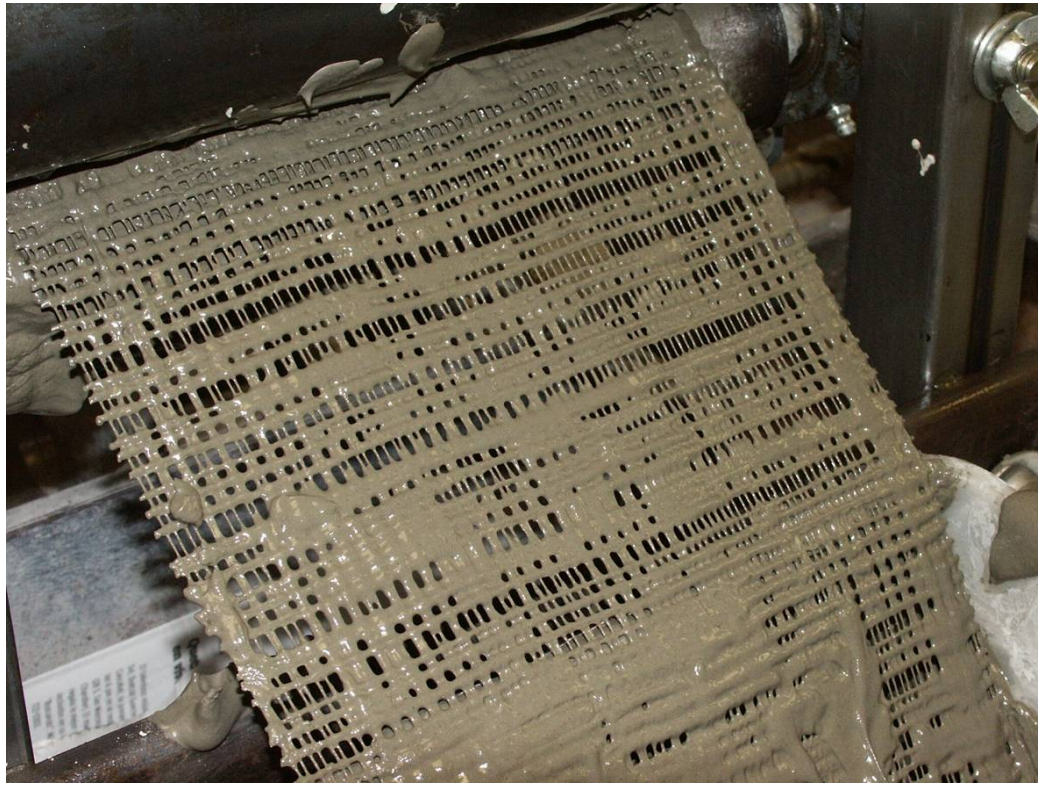
Concrete used worldwide, Tensile Strength = 2 MPa, strain capacity = 0.02%
Pultruded Cement composites, Tensile Strength = 50 MPa, strain capacity = 1%



Pultrusion of Textile Composites

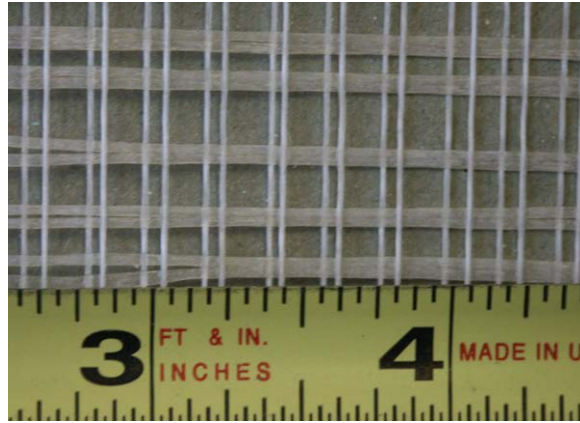


Slurry Infiltrated Fabric

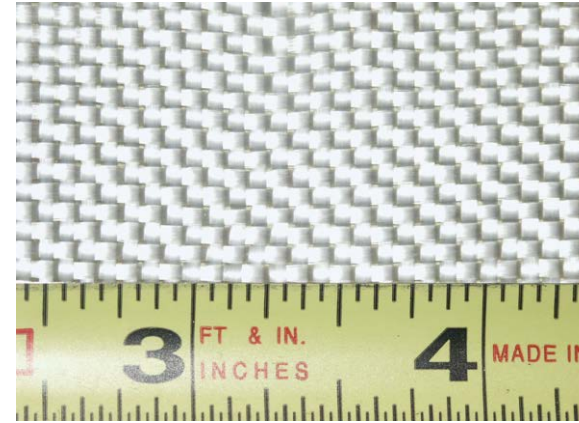


Mobasher, B., and Pivacek, A., "A Filament Winding Technique for Manufacturing Cement Based Cross-Ply Laminates," *Journal of Cement and Concrete Composites*, 20 (1998) 405-415.

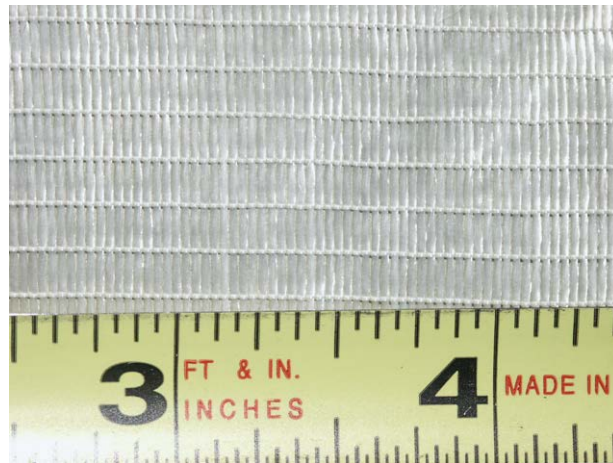
Typical textile Types



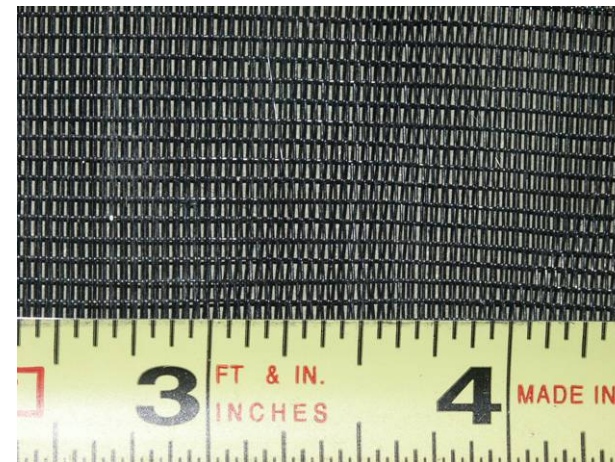
AR-Glass



PVA

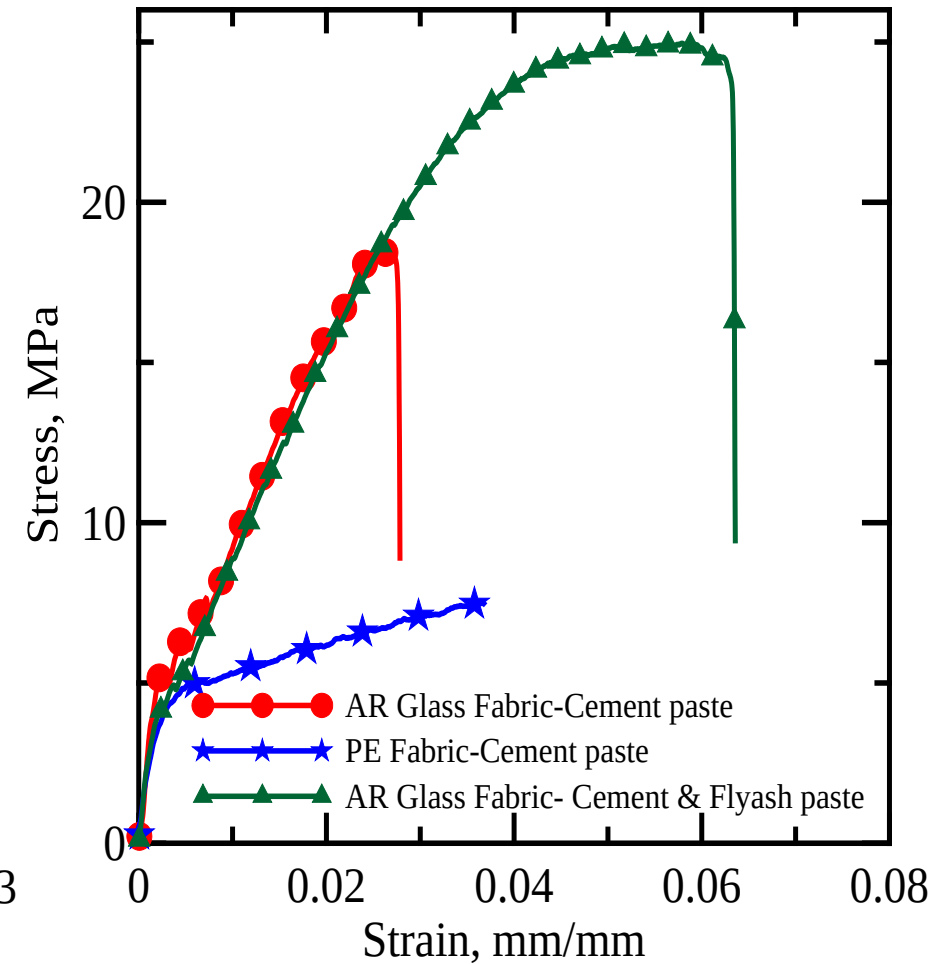
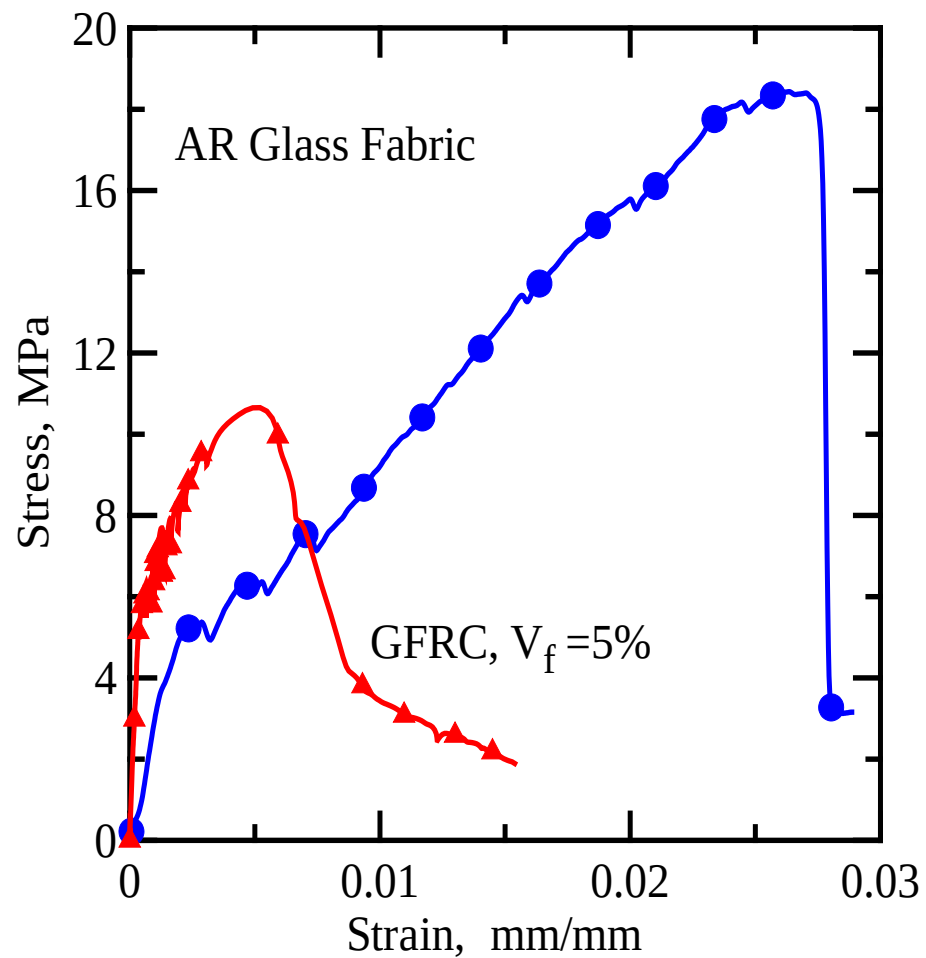


PP

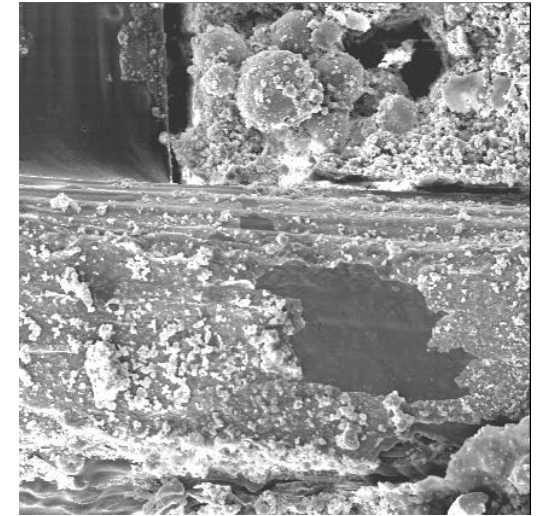
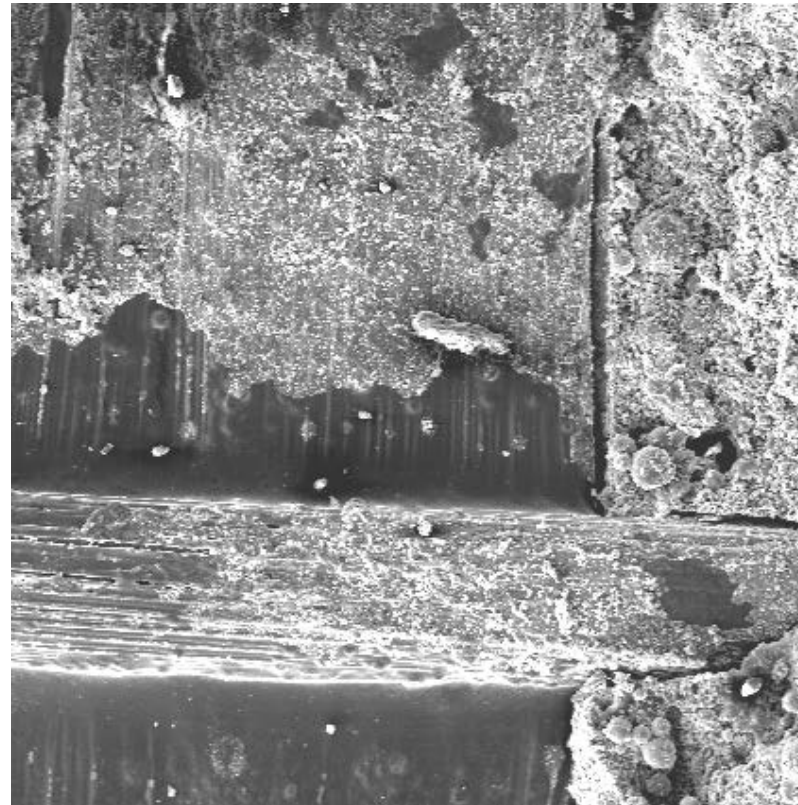
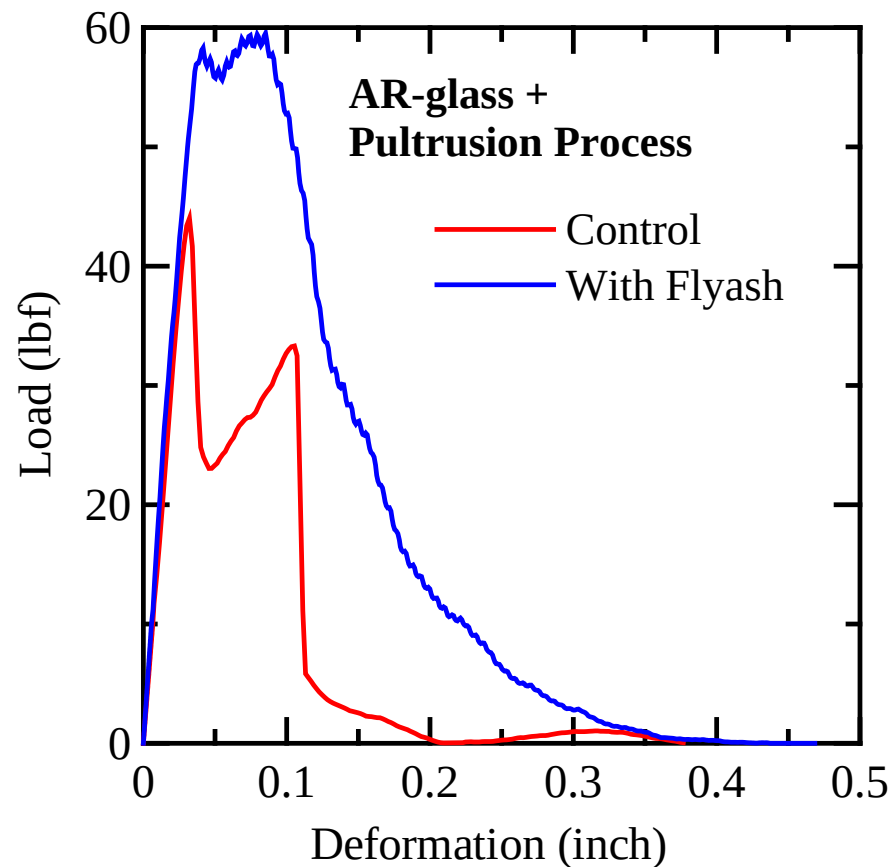


PE

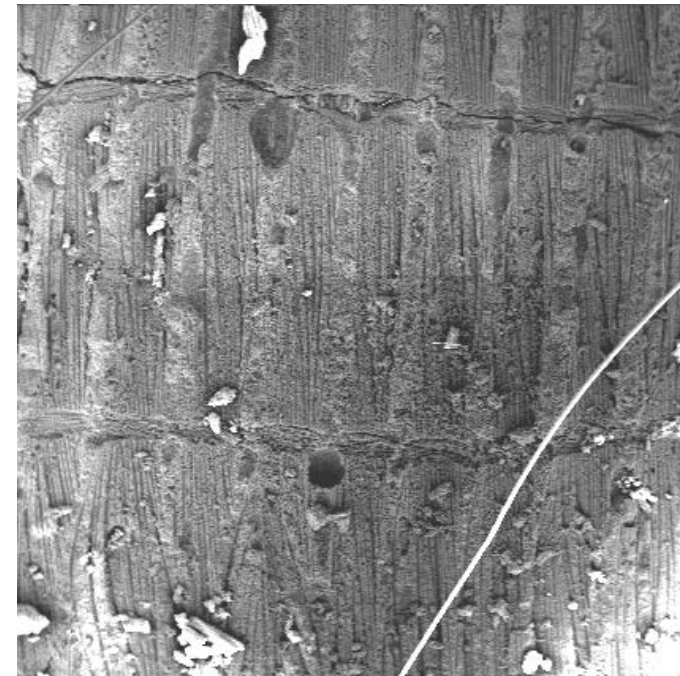
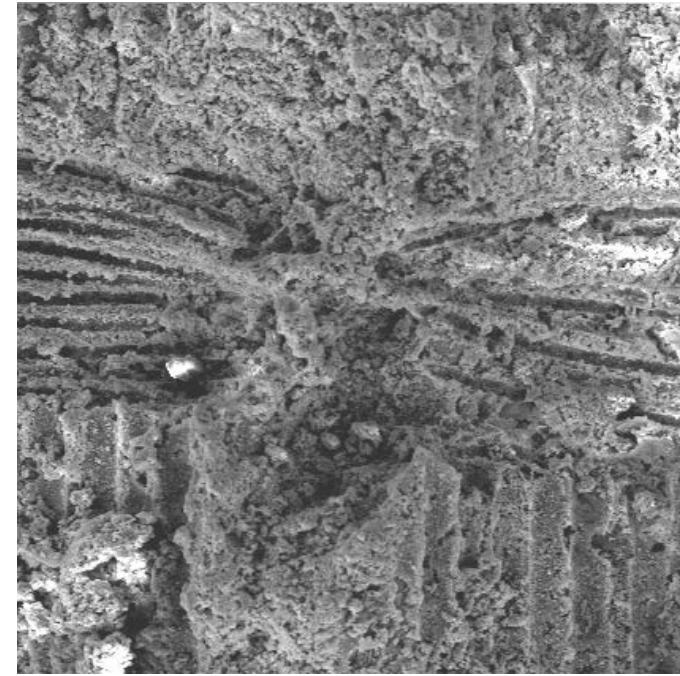
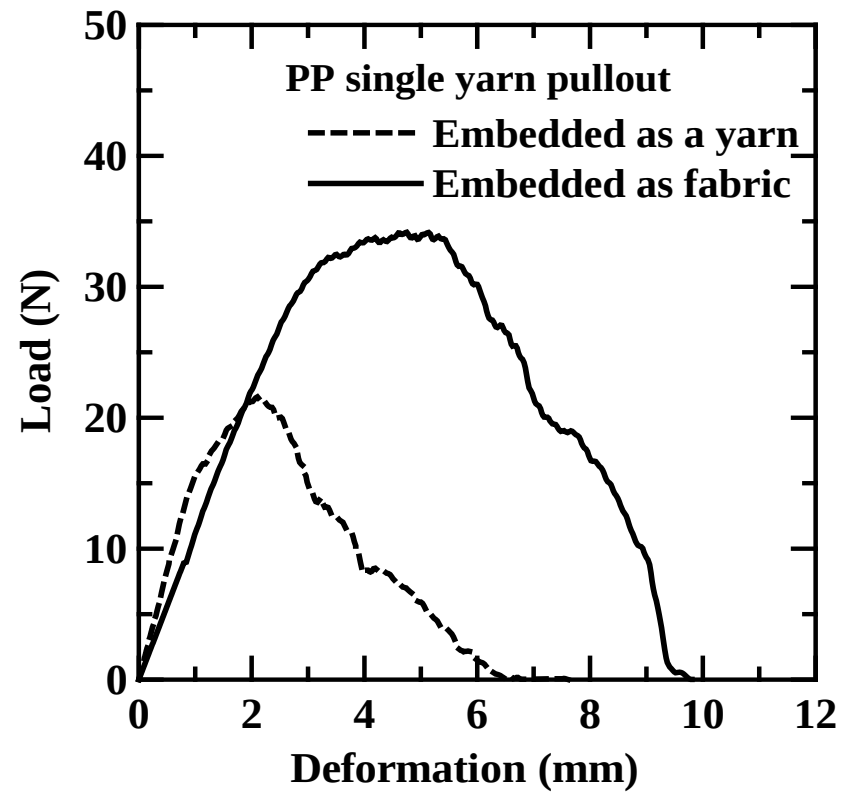
Comparison of Properties with conventional high fiber cement-based composites



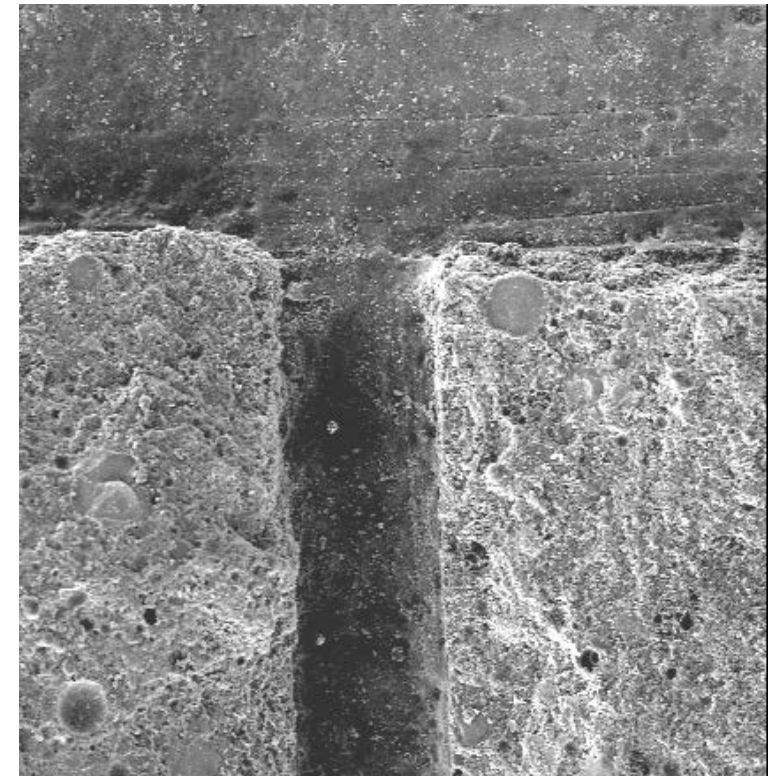
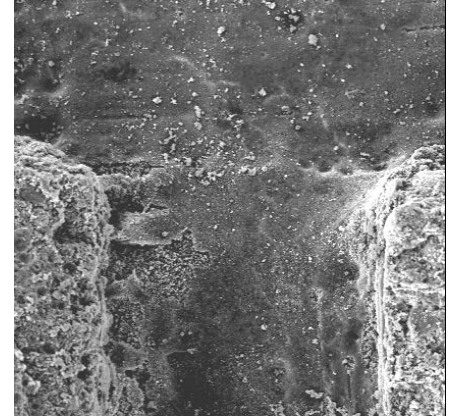
Effects of Matrix formulation and densification using Flyash on the overall bond response



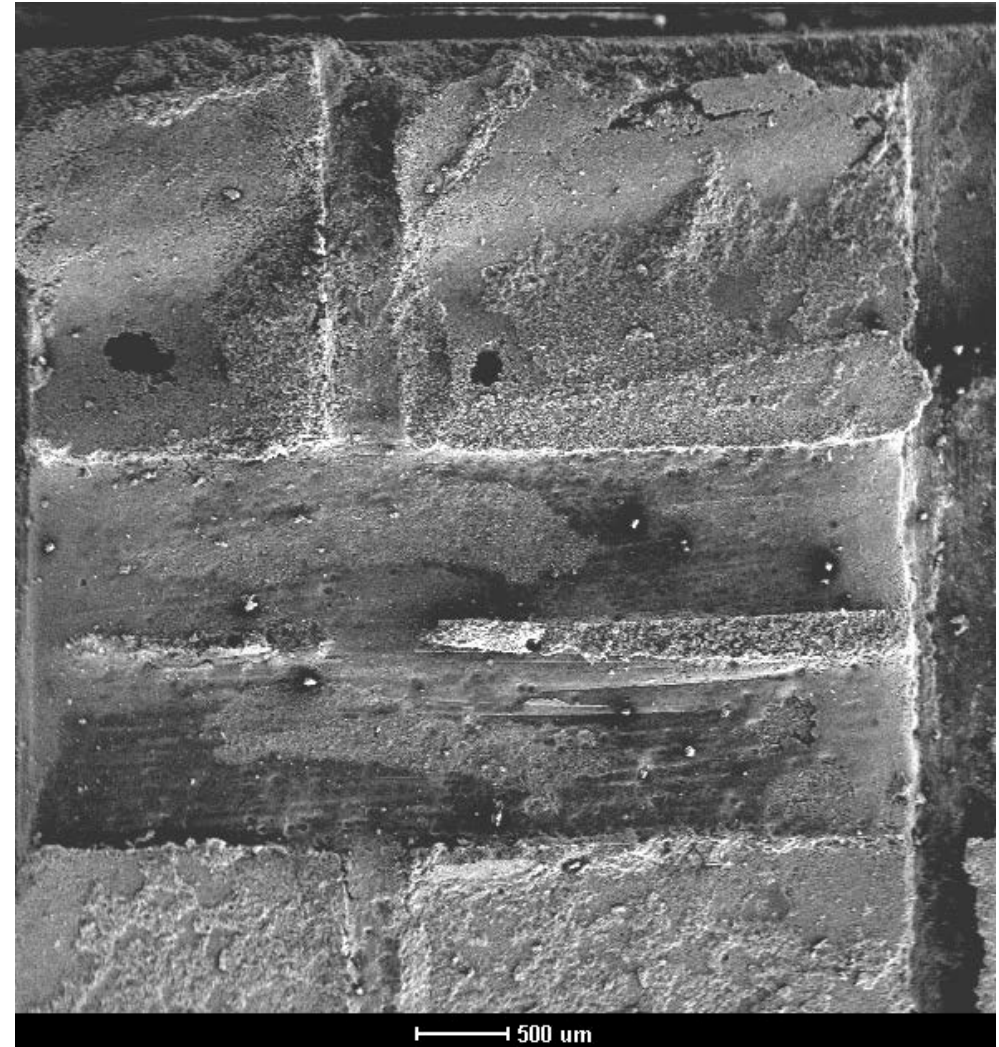
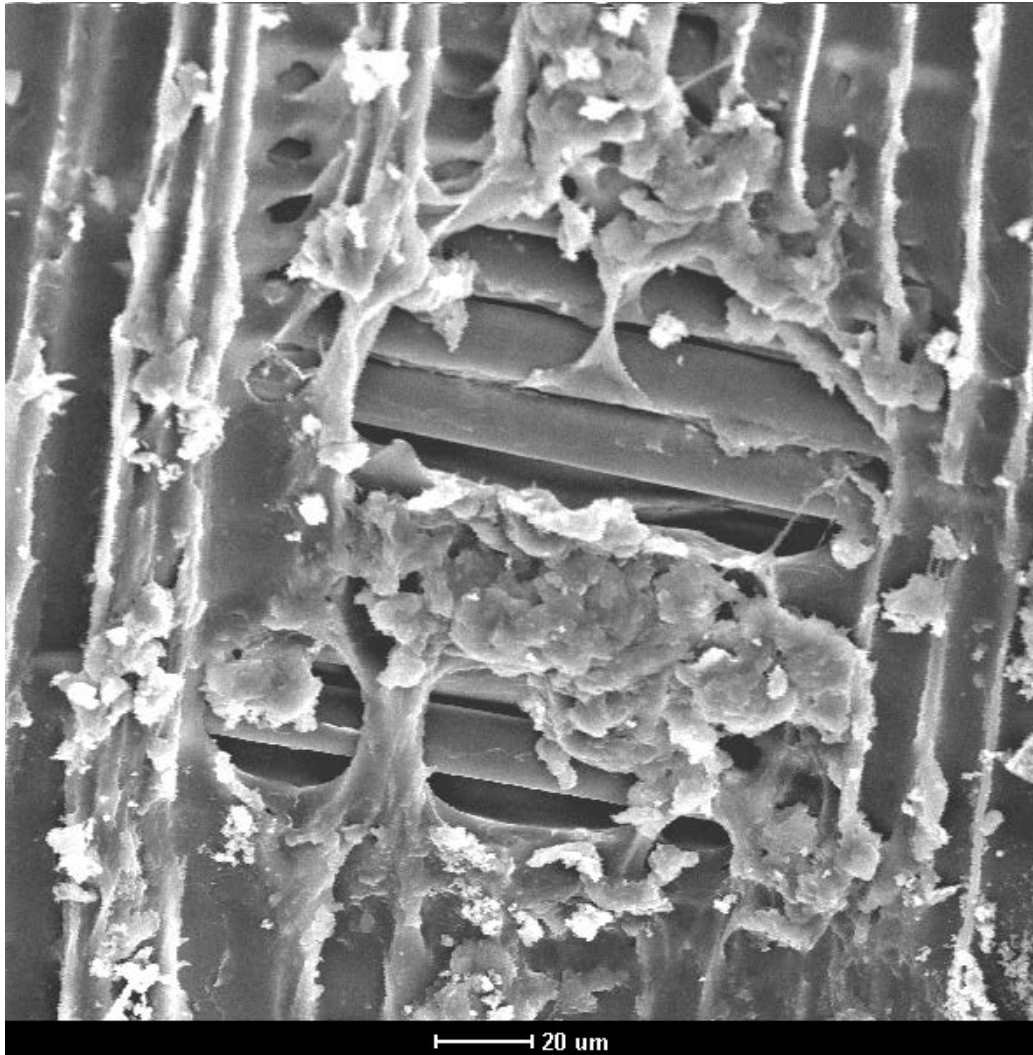
Effects of anchorage on the overall pullout response



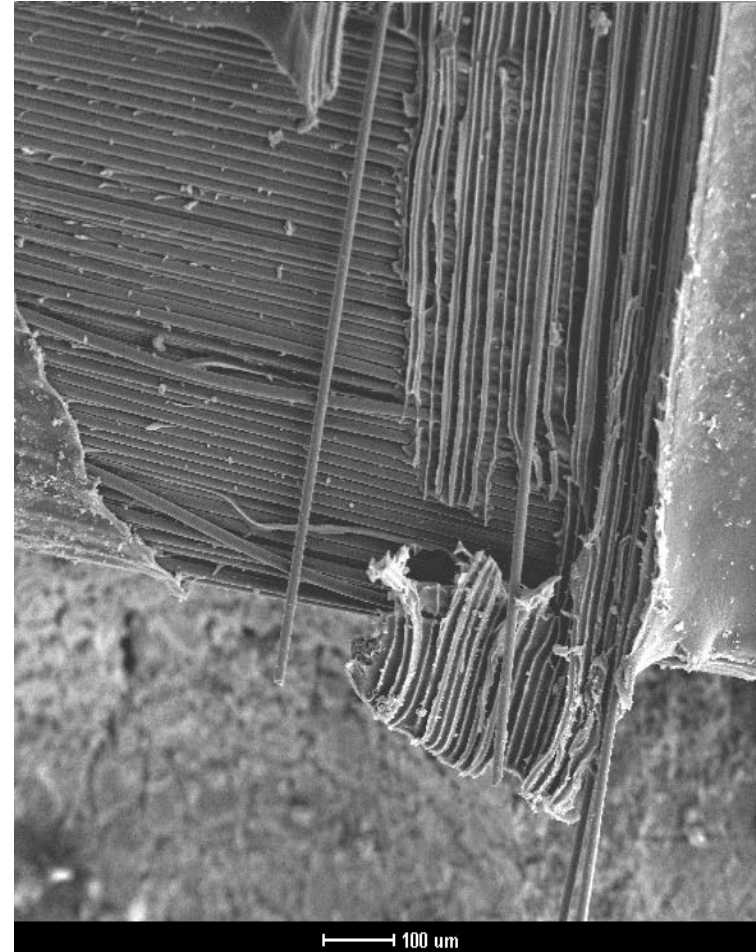
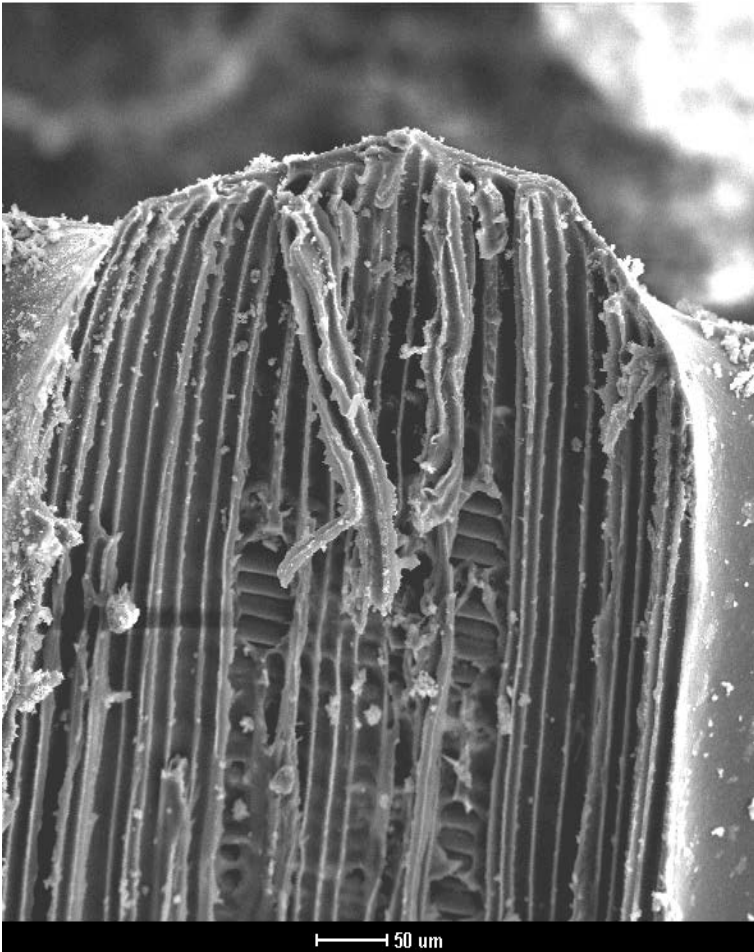
Anchorage due to interlock



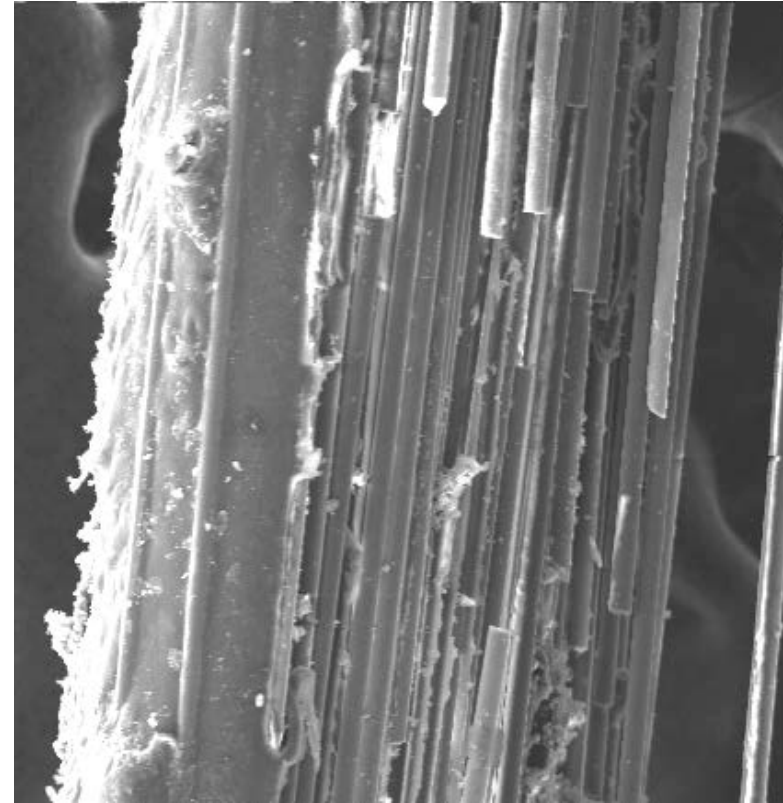
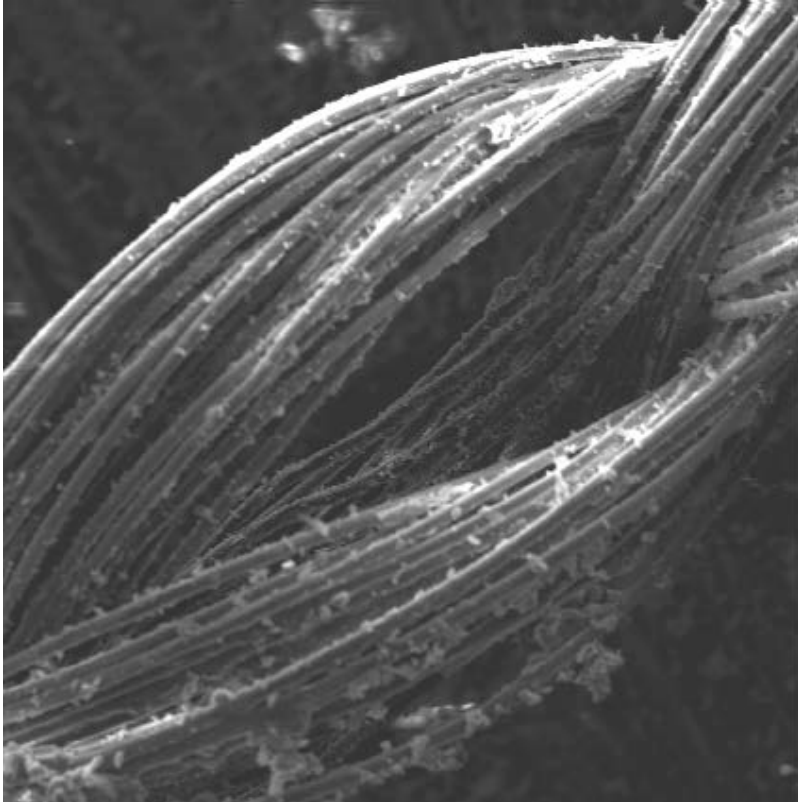
Load Transfer Mechanism Attributed to the Failure of Junctions



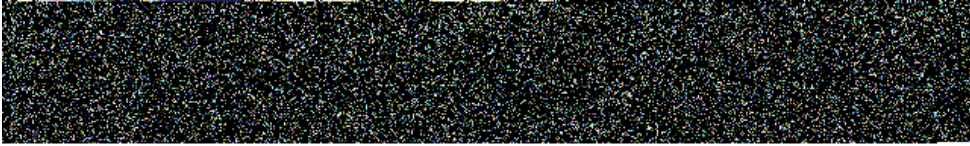
Failure of Junctions



High efficiency factor obtained by anchorage of polymeric & fracture of brittle fibers

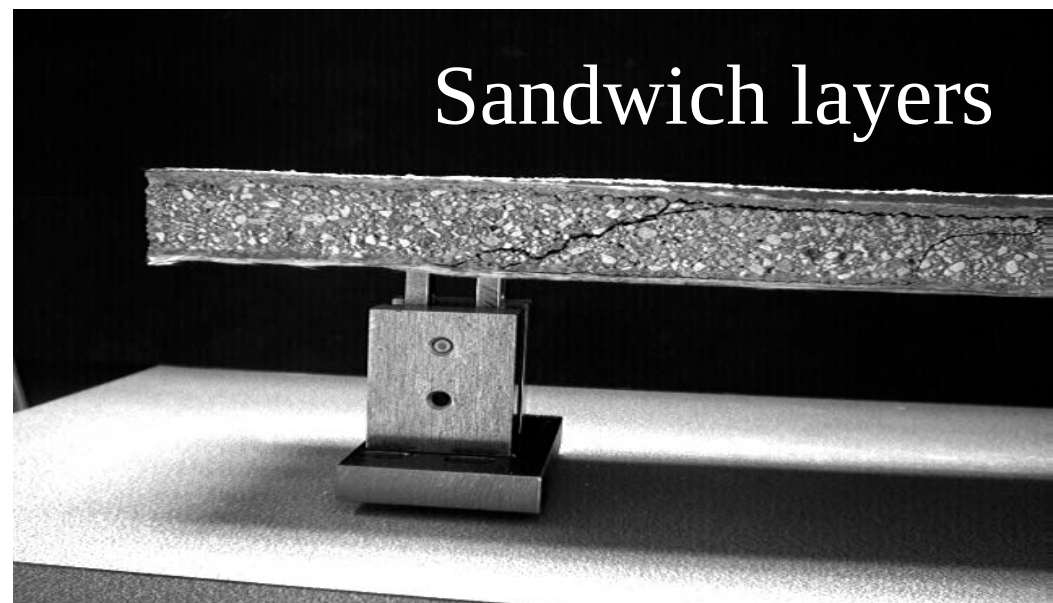
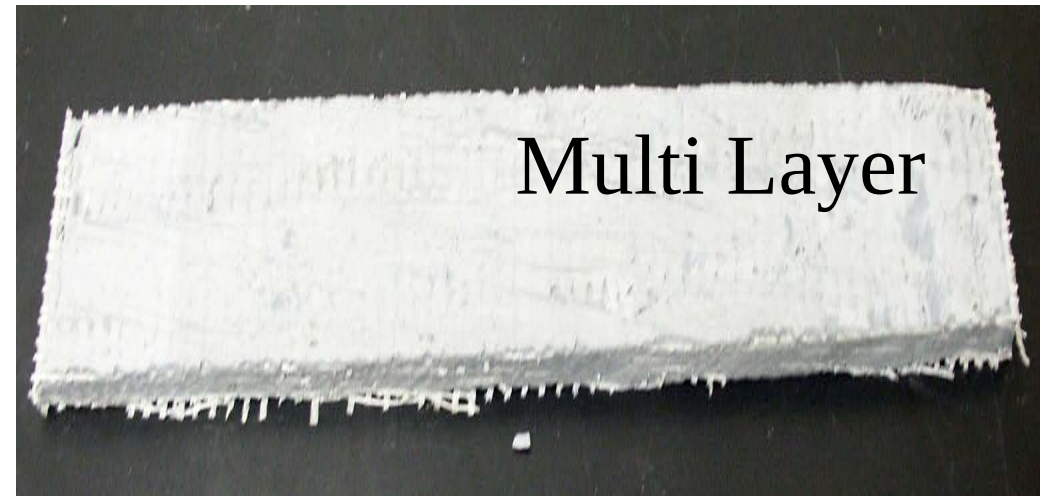


Macroscopic Crack Distribution

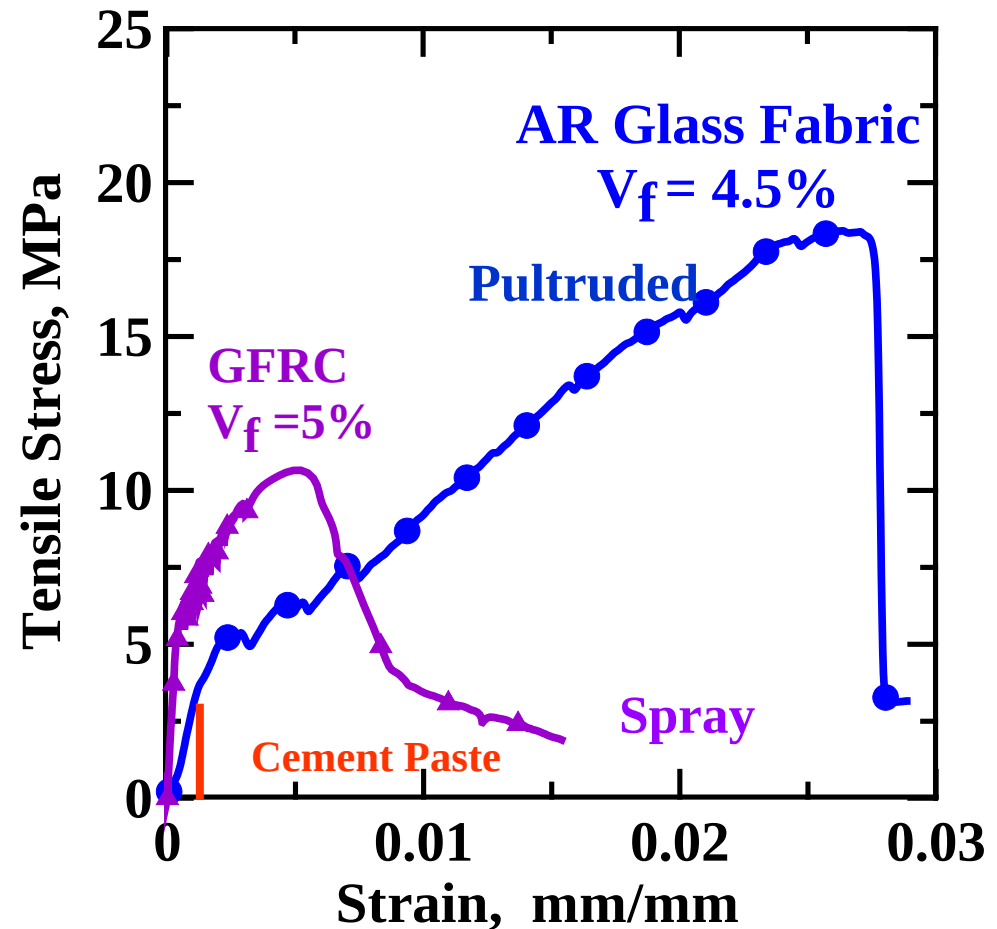


New composites with fabrics

- The special geometry of the yarn in fabric
- Mechanical anchoring of the individual yarn in the fabric with the cement matrix
- Important when using low modulus yarns

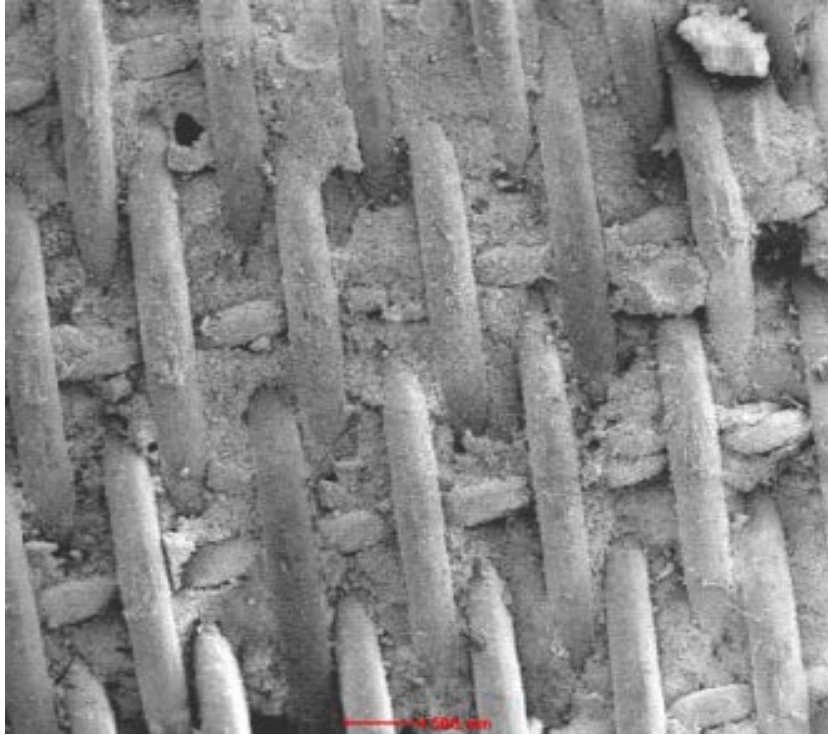


Effect of Processing- Fabrics vs. Conventional GFRC

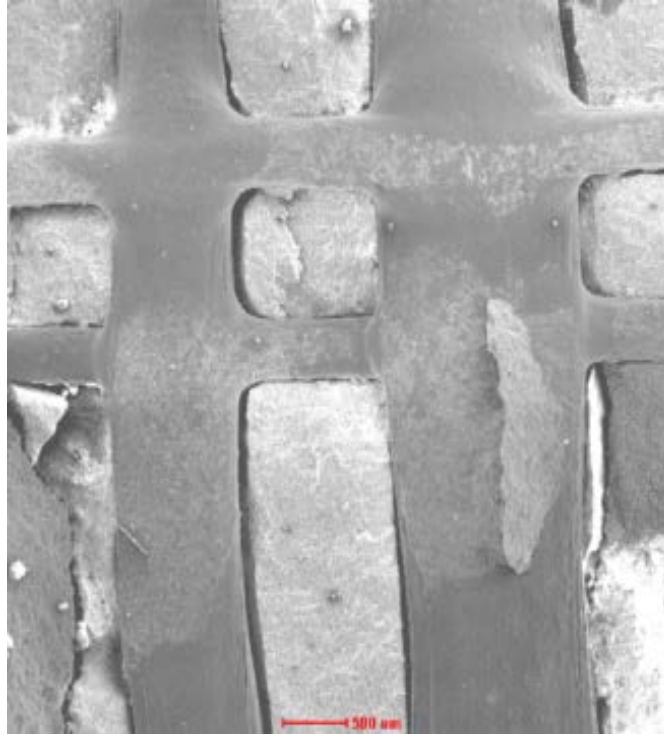


A. Peled, and B. Mobasher, "Cement Based Pultruded Composites with Fabrics," Proceedings of the 7th International Symposium on Brittle Matrix Composites (BMC7) in Warsaw, October 13-15, 2003.

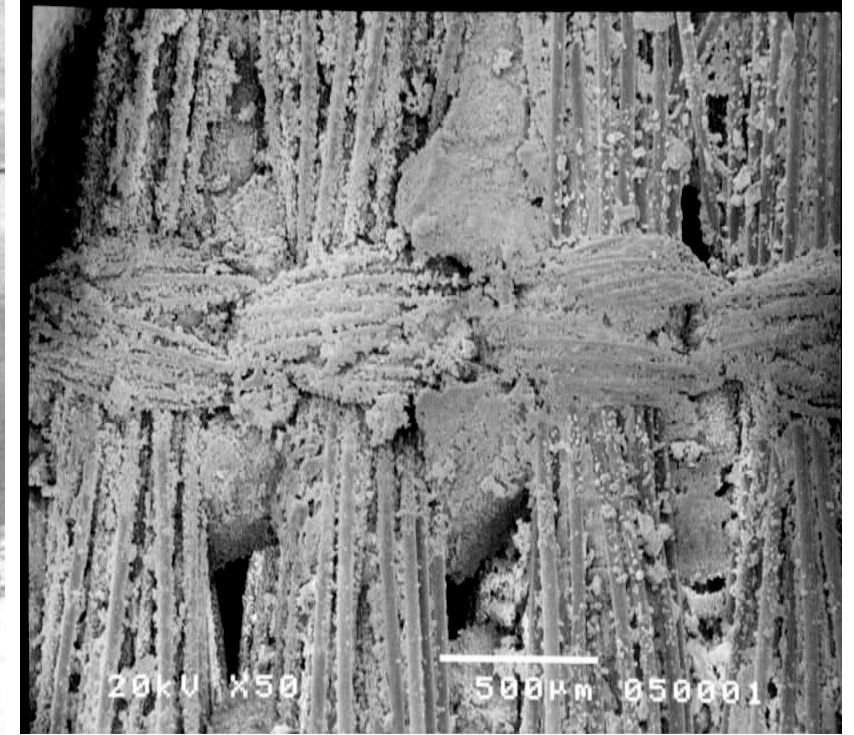
Impregnation of textiles



Polyethylene
(PE)
Woven Fabric
 $E=2$ GPa

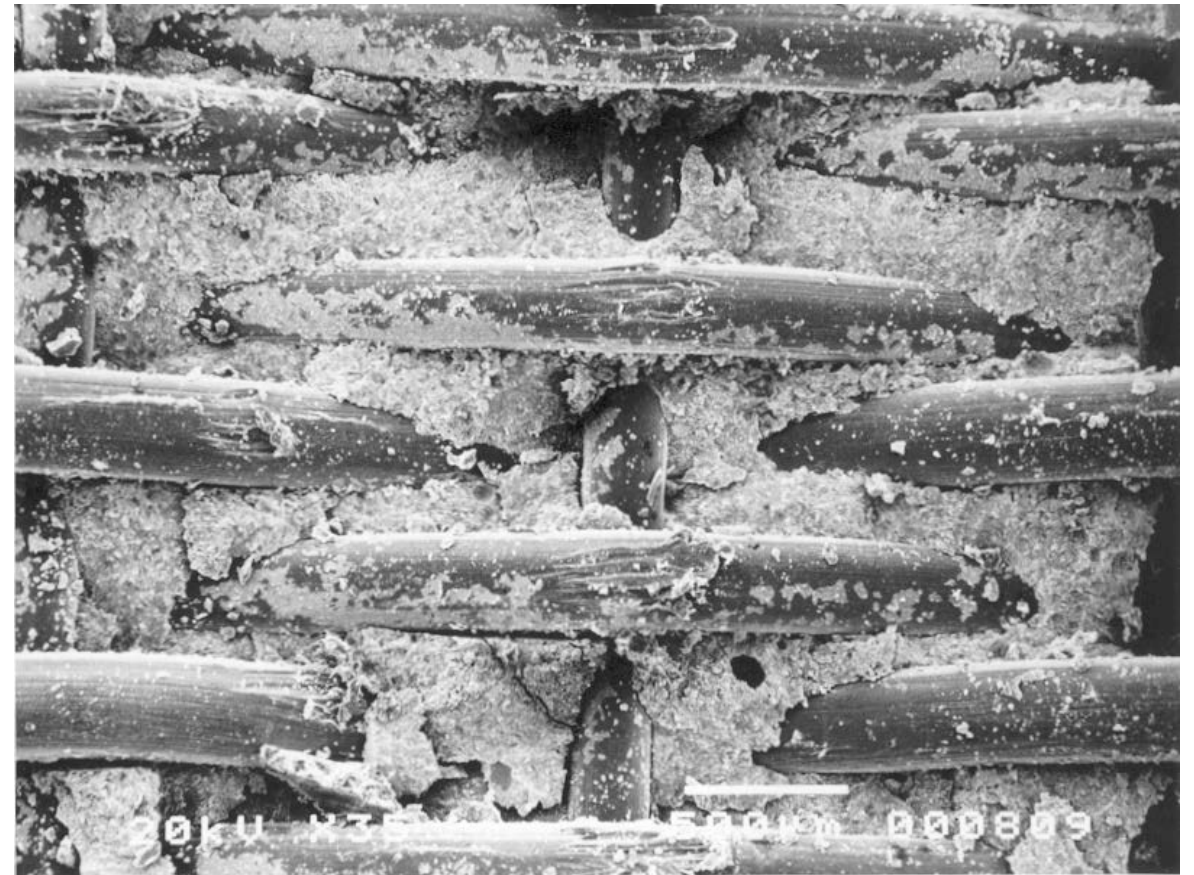


AR Glass
Bonded Fabric
 $E=78$ GPa

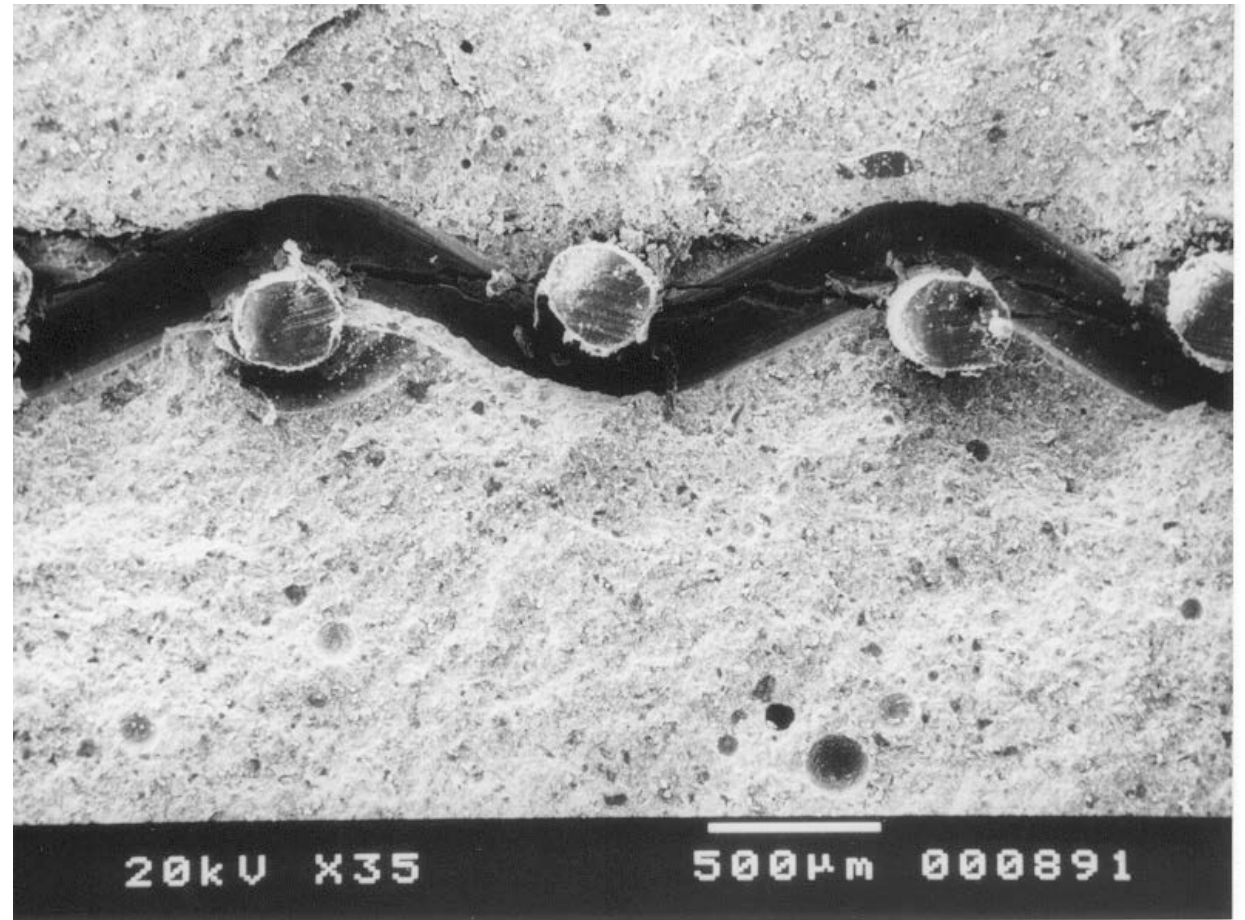


Polypropylene (PP)
Knitted Fabric
 $E=6$ GPa

Open net mesh allows penetration between the fabric openings and mechanical anchoring



- Enable to control: orientation, distribution and volume content of the reinforcement
- Different geometries of fabrics - wide variety of properties
- Easy to handle and place in precast products
- Cheap polymeric fabrics, PP, PVA, PE

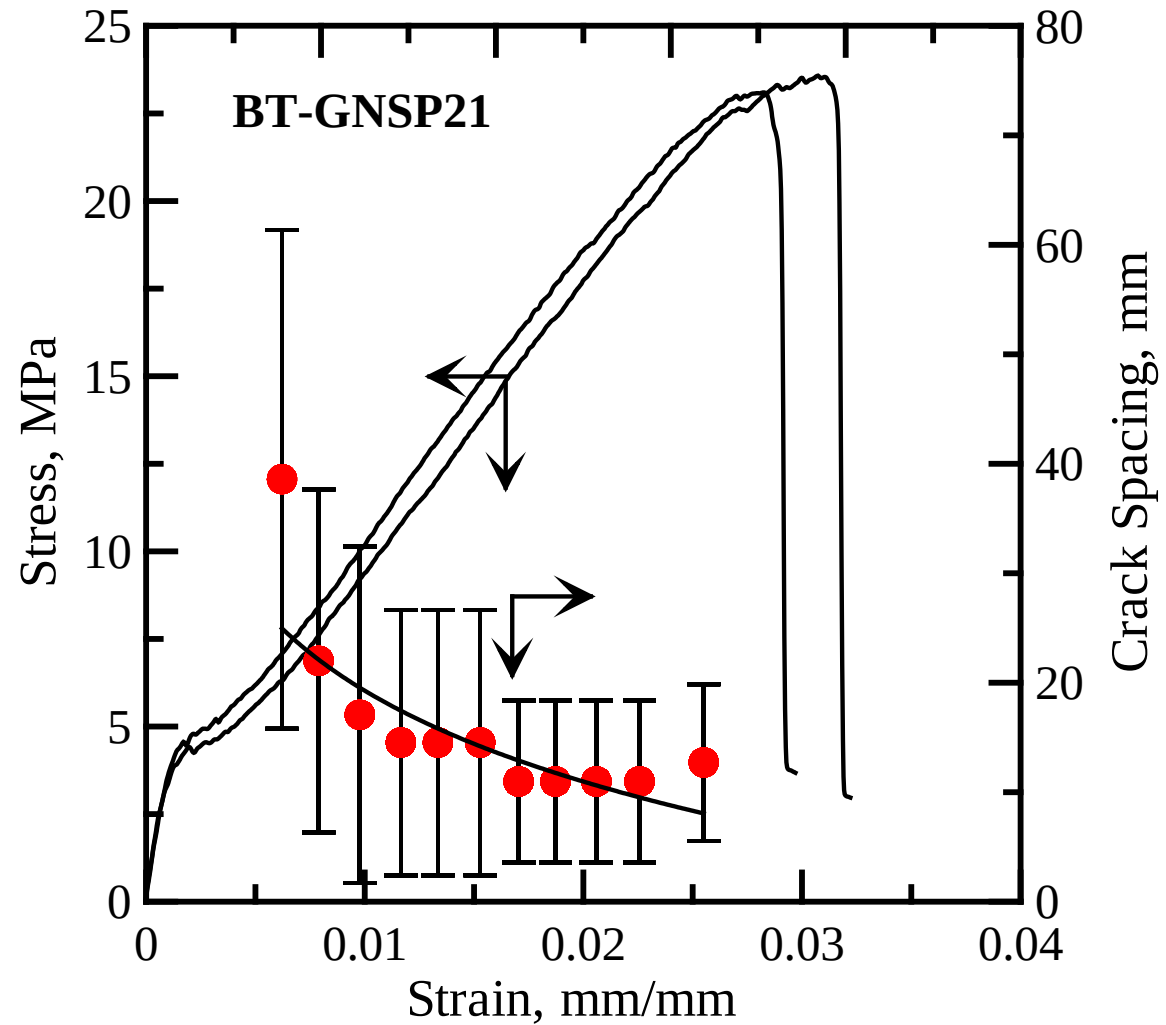


Anchorage of polymeric based fabrics is the primary advantage for fabrics as reinforcement

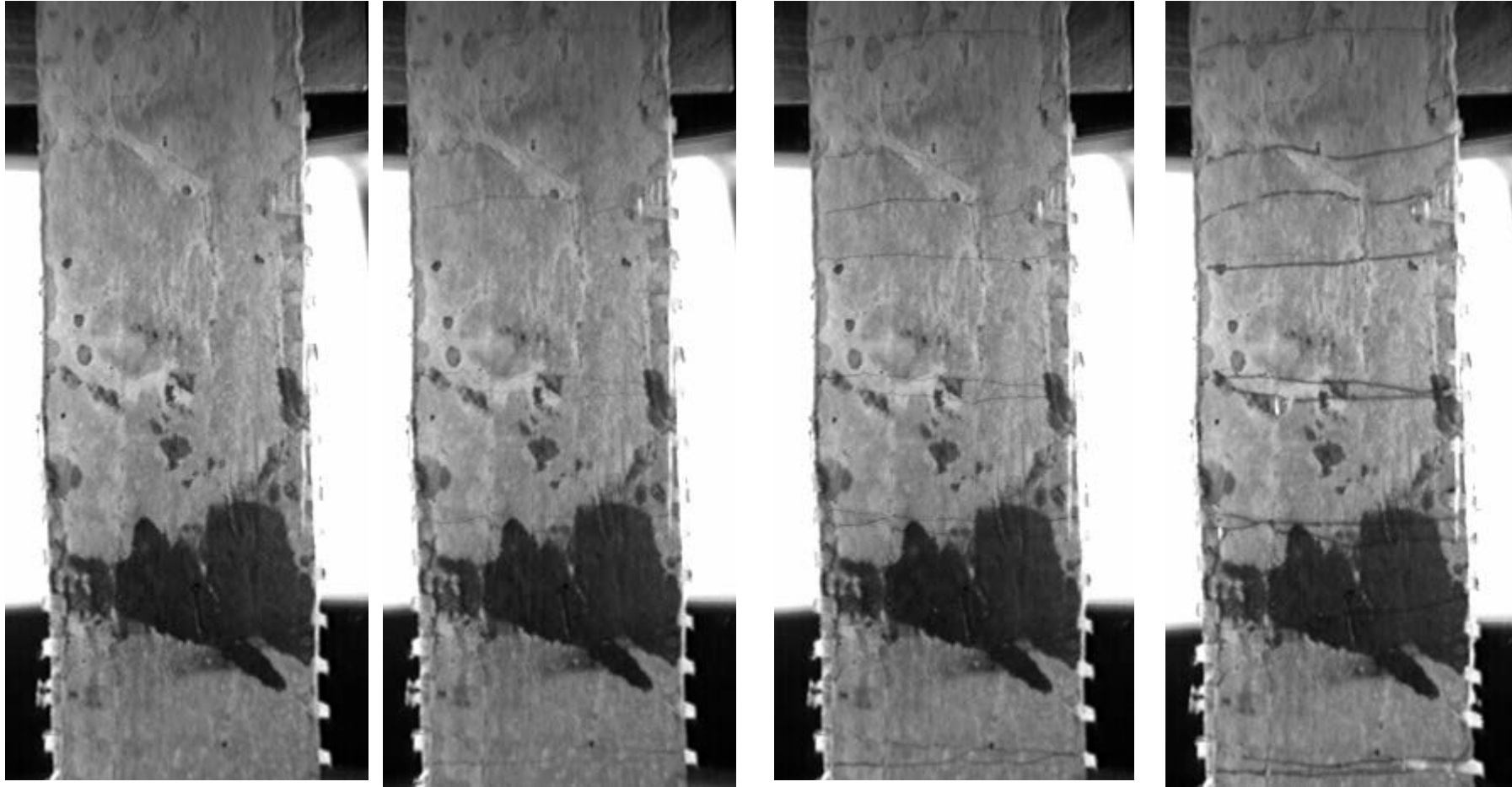
Bonded Fabrics- Alkali Resistant Glass



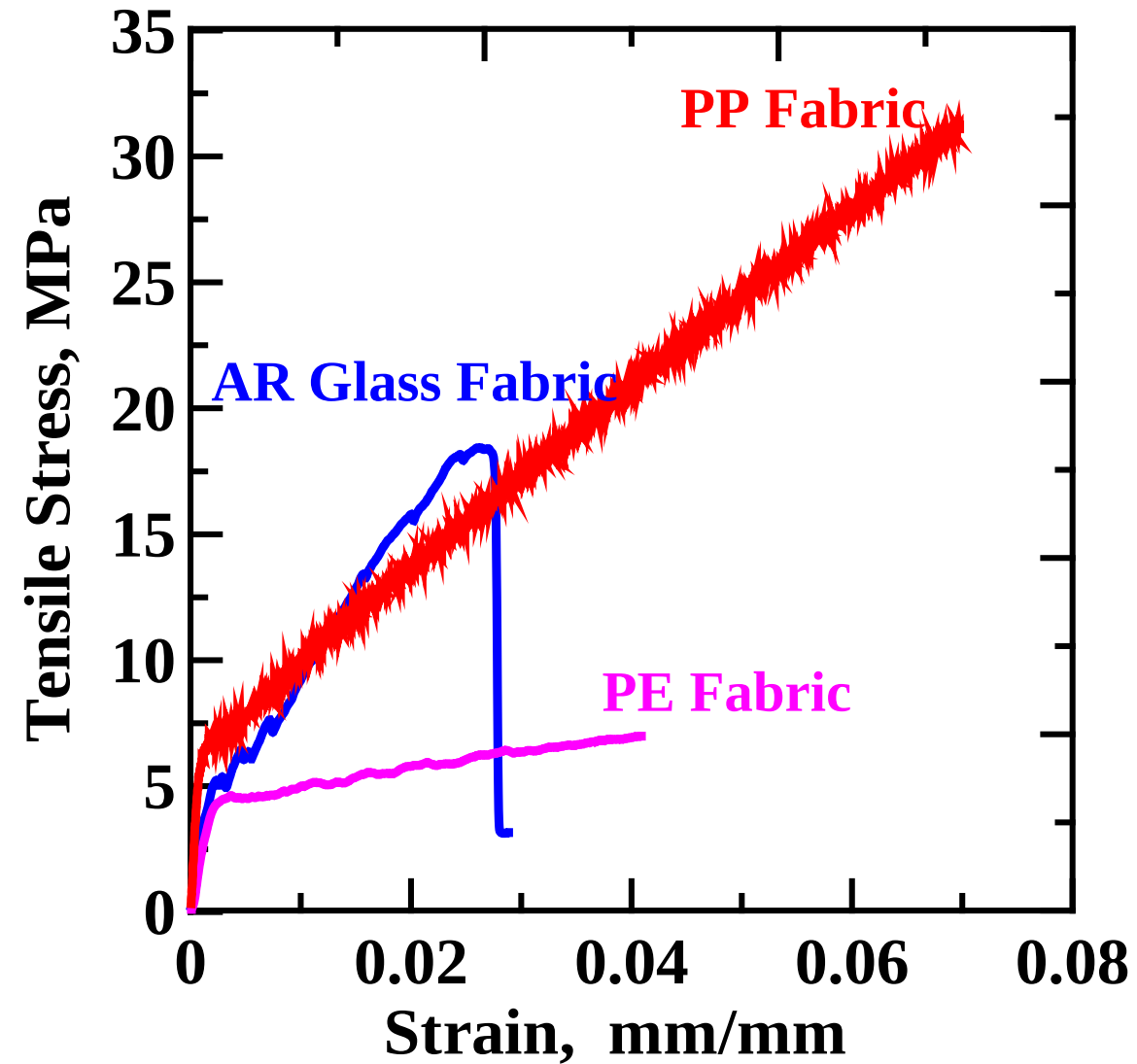
Tensile Stress strain response and Evolution of Crack Spacing



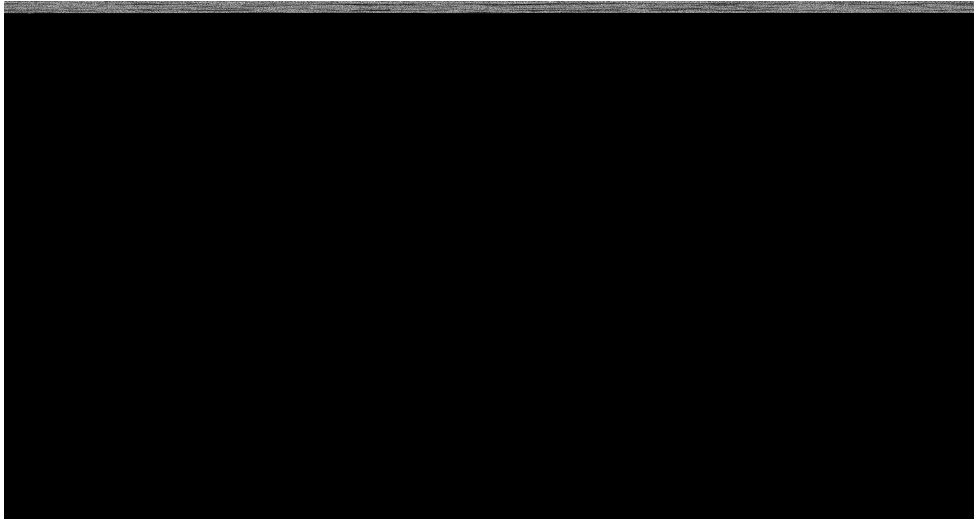
Various stages of cracking



Tensile Behavior-Effect of Fabric Type



Showcase Studies



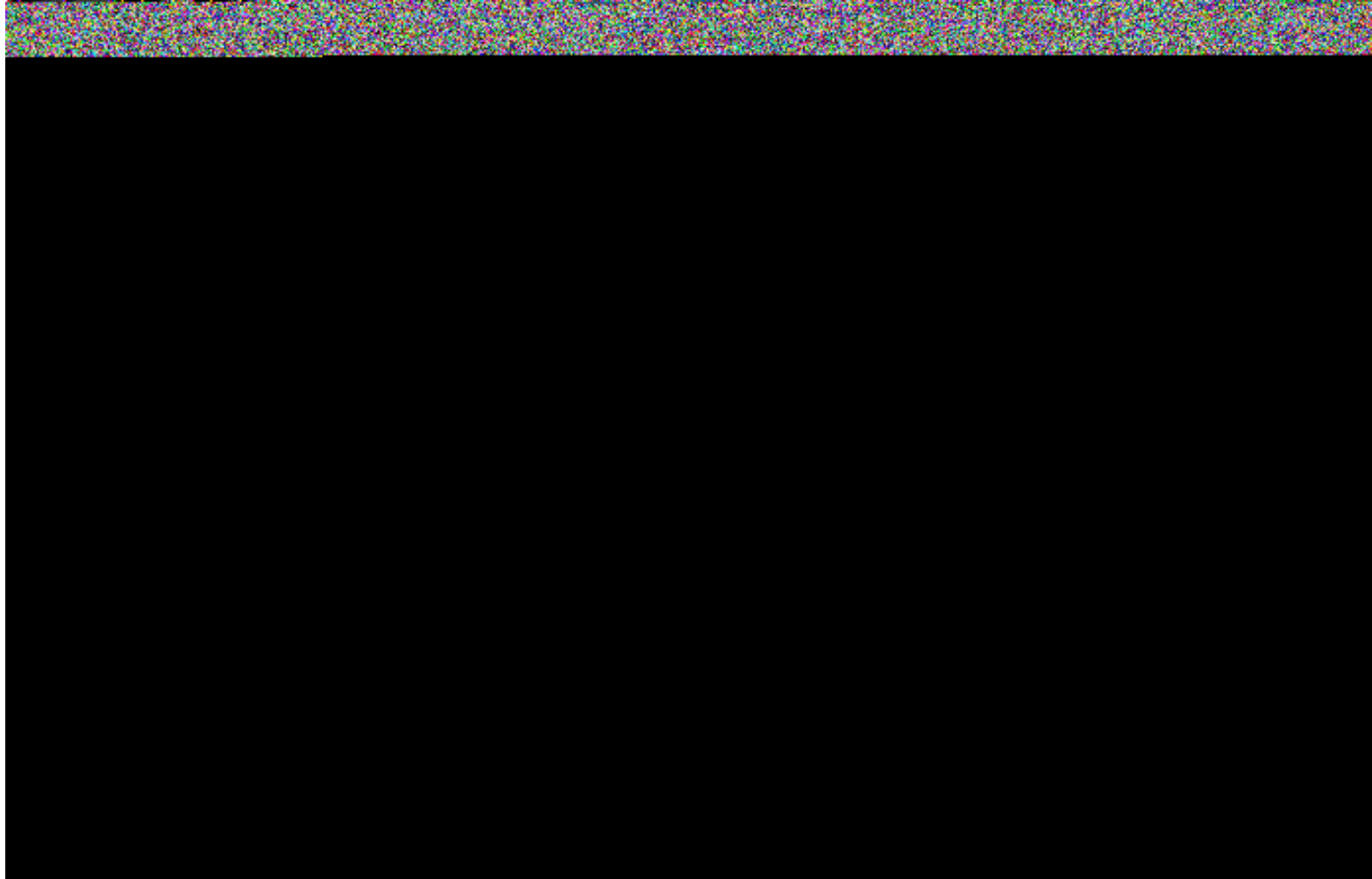
U-shaped cross section of the textile reinforced concrete bridge in Oschatz



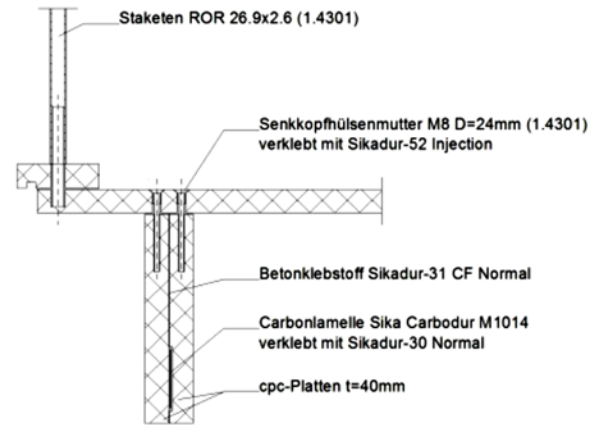
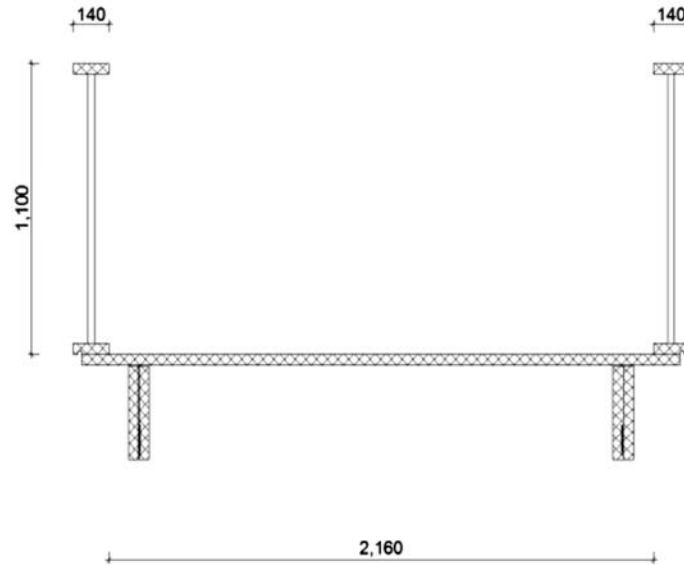
Textile reinforced concrete bridge in Kempten, Photo: Harald Michler

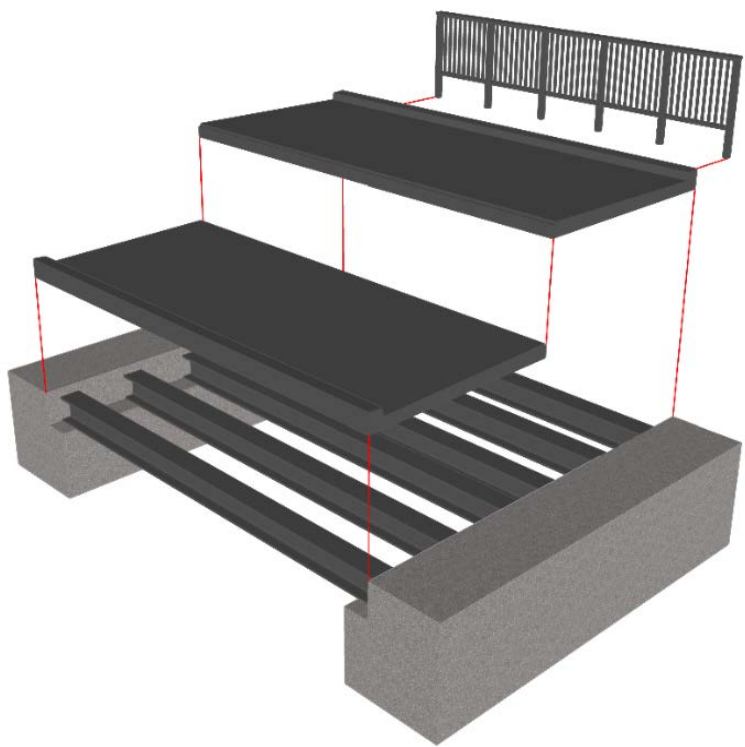


Textile reinforced concrete pedestrian bridge with Multiple web cross section of in Albstadt-Lautlingen in Albstadt-Lautlingen,

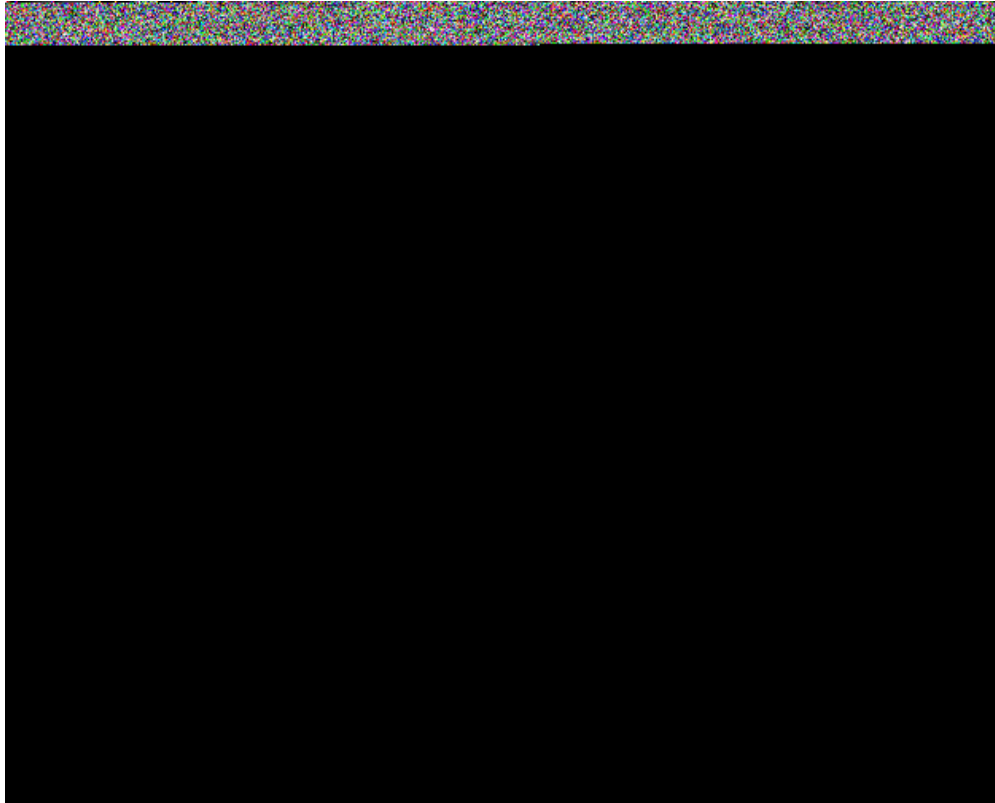


Textile reinforced concrete bridge in Winterthur





Repair and Retrofit



Bridge in Naila rehabilitate with textile reinforced concrete, Photo: Frank Schladitz

Textile reinforced concrete cladding of the Bosphorus Bridge in Istanbul in Turkey,
Photo: Fibrobeton

